1 Solution — GIAM 1.3 Problem 3

1.1 Problem

The sentence “For every pair of (distinct) real numbers there is another real number between them.” is true. Why?

1.2 The Short Answer

Given any two distinct reals, their **midpoint** lies strictly between them:

$$\boxed{\text{For distinct } a, b \in \mathbb{R}, \text{ the midpoint } m = \frac{a+b}{2} \text{ satisfies } a < m < b \text{ (taking } a < b\text{).}}$$

Because the midpoint is itself a real number and works for *every* pair, the universal statement is true. ■

1.3 Proof

The sentence is a universally quantified statement with an existential inside — a "∀∃” claim:

$$\forall a\, \forall b \left(a \neq b \;\to\; \exists m \in \mathbb{R}\; (m \text{ is strictly between } a \text{ and } b)\right).$$

To prove it, take *arbitrary* distinct reals a, b. Without loss of generality $a < b$ (otherwise rename them). Define

$$m \;=\; \frac{a+b}{2}.$$

This m is a real number (the reals are closed under addition and division by 2). Check it lies strictly between a and b:

$$m - a = \frac{a+b}{2} - a = \frac{b-a}{2} > 0 \quad (\text{since } b > a) \implies m > a,$$

$$b - m = b - \frac{a+b}{2} = \frac{b-a}{2} > 0 \implies m < b.$$

Hence $a < m < b$. Since a, b were an arbitrary distinct pair, *every* pair has a real number between them. ■

1.4 Picture — Density of the Reals

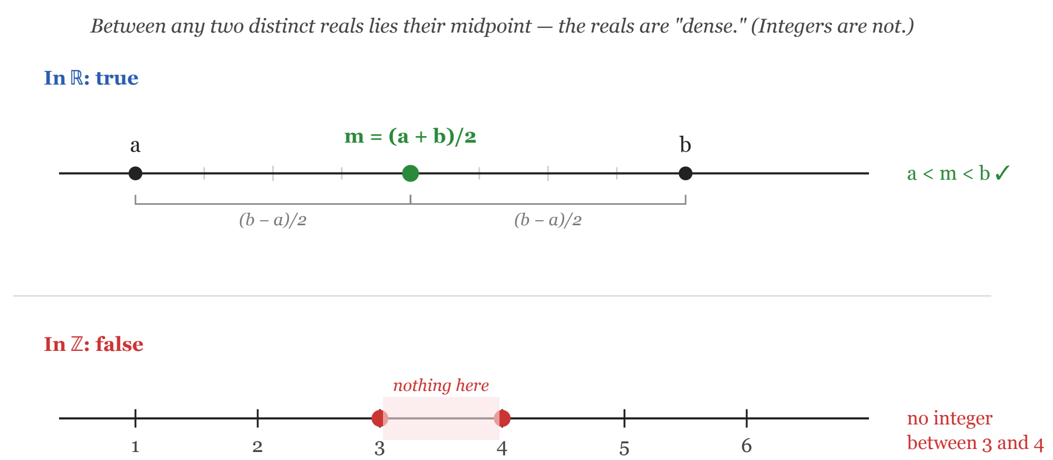

Figure 1.1: Top: for reals a < b, the midpoint m = (a+b)/2 sits exactly halfway between them, at distance (b−a)/2 from each — so a < m < b. The faint ticks hint that the process repeats forever (infinitely many reals between any two). Bottom: in the integers, nothing lies strictly between 3 and 4, so the same sentence is false for ℤ.

1.5 Remark — This Is "Density"

The property just proved has a name: the real numbers are **dense** (more precisely, $\mathbb{R}$ is a *densely ordered* set). Density means: between any two distinct elements there is always a third. The midpoint construction is the cleanest possible witness — though far from the only one (see below).

1.6 Remark — Proving "∀" Needs a Construction, Not Examples

This problem is the mirror image of Problem 2. There, we *disproved* a universal statement ("every metal is solid") with a single counterexample. Here we must *prove* a universal statement is true — and for that, one example is *not* enough. Checking that 3.5 lies between 3 and 4 would verify a single instance, not the claim "for *every* pair."

Proving a "$\forall$" requires a method that works for an *arbitrary* input. The midpoint formula $m = (a+b)/2$ is exactly such a method: it takes *any* distinct a, b and produces a valid between-number. This is the asymmetry from Problem 2 seen from the other side — **disproving "all" needs one case; proving "all" needs a general argument.**

1.7 Remark — The Witness Depends on the Inputs (∀∃ Structure)

The statement is "$\forall\exists$": *for all* a, b, *there exists* m. Crucially, the m that exists **depends on** a **and** b — it is not one fixed number that works for every pair. The dependence is captured exactly by the midpoint *function* $m(a,b) = (a+b)/2$ (a so-called *Skolem function* for the existential quantifier).

This is the order-of-quantifiers point the section emphasises. The true reading is

$$\forall a\, \forall b\, \exists m\, (a < m < b),$$

with $\exists m$ *inside*. Swapping to put $\exists m$ *outside* —

$$\exists m\, \forall a\, \forall b\, (a < m < b)$$

— would assert that *one single* real number lies between *every* pair, which is absurd. The truth lives entirely in the $\forall\exists$ ordering.

1.8 Remark — Not Just One Between, but Infinitely Many

The sentence only asks for "another" real — i.e. *at least one*. But the midpoint trick can be *iterated*: between a and m there is *their* midpoint, and between that and a another, and so on without end. So in fact there are **infinitely many** reals between any two distinct reals — uncountably many, in truth, since the whole interval (a, b) is squeezed in there. The faint repeated ticks in the figure hint at this endless subdivision.

A vivid consequence: **no two distinct reals are ever "adjacent."** Unlike the integers, where 4 is the immediate successor of 3, a real number has *no* "next" real — there is always room for more in between.

1.9 Remark — Why the Universe of Discourse Is Essential

The truth of this sentence hinges entirely on what set the numbers come from — the recurring lesson of these sections:

Universe	**"between any two distinct elements lies another"?**
$\mathbb{Z}$	**false** — nothing lies between 3 and 4
$\mathbb{Q}$	**true** — the midpoint of two rationals is rational
$\mathbb{R}$	**true** — the midpoint of two reals is real

Table 1.1

The integers are *discrete*: consecutive integers have a gap with nothing inside, so the statement fails (the bottom panel of the figure). The rationals and reals are *dense*, and the midpoint argument works verbatim in both — note $(a + b)/2 \in \mathbb{Q}$ whenever $a, b \in \mathbb{Q}$. As always, a quantified statement is only meaningful once its universe is fixed; the *same words* are true over $\mathbb{R}$ and false over $\mathbb{Z}$.

1.10 Remark — The Midpoint Is Not the Only Witness

Any number strictly inside the interval works — the midpoint is just the symmetric, memorable choice. For distinct $a < b$, every **convex combination**

$$m_\lambda \;=\; (1 - \lambda)\, a + \lambda\, b, \qquad 0 < \lambda < 1,$$

satisfies $a < m_\lambda < b$ (the midpoint is the case $\lambda = \frac{1}{2}$). As λ ranges over the open interval $(0, 1)$ — itself uncountably many values — we get uncountably many distinct between-numbers, re-deriving the "infinitely many" remark. The freedom to pick *any* λ underscores that the existential witness is abundant, not scarce: density is not a near-miss property but an overwhelmingly satisfied one.